👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 3 4-Stroke Theory & Engine Parts (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
His readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as his worksheets for that lesson
- Quick check: score, then scale to his test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **iCADEMY Bronze 6**: STIHL iCADEMY is login-only; we wrote our own engine-principles content.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

⚙️ 4-Stroke Theory & Engine Parts: Guided Notes

Small Engines Overhaul · Unit 3 · Level 3 (closer to the class work) · goes with the 4 Stroke Theory slides, Parts ID quiz and Engine Tools/Parts sheets

Name: _____

Use the class slides and your reading. Key words:

stroke *intake* *compression* *power* *exhaust* *piston* *connecting rod* *crankshaft* *camshaft* *flywheel* *two-stroke*

A. Strokes

1. Fill in the table.

Stroke	Piston direction	Valves
Intake		
Compression		
Power		
Exhaust		

B. Parts

2. The _____ links the piston to the crankshaft.

3. The _____ opens the valves.

4. The _____ keeps the engine turning between power strokes.

C. Compare

5. Why do chainsaws use 2-stroke engines?

_____

_____

⚙️ 4-Stroke Theory & Engine Parts: Reading

Small Engines Overhaul · Unit 3 · Level 3 (closer to the class work) · goes with the 4 Stroke Theory slides, Parts ID quiz and Engine Tools/Parts sheets

Name: _____

How a Four-Stroke Engine Works

A small gas engine turns the energy in gasoline into motion. Most lawn mowers and many other small engines use a four-stroke cycle. A stroke is one movement of the piston, up or down, inside the cylinder.

On the intake stroke, the piston moves down and the intake valve opens, pulling in a mixture of air and fuel. On the compression stroke, both valves close and the piston moves up, squeezing the mixture. On the power stroke, the spark plug fires, the mixture burns and expands, and the piston is forced down. On the exhaust stroke, the exhaust valve opens and the piston pushes the burned gases out.

The piston is connected to the crankshaft by the connecting rod. The crankshaft changes up-and-down motion into rotation. It turns twice for every four strokes, and the camshaft, turning at half speed, opens each valve at the right moment. A heavy flywheel keeps the engine turning between power strokes.

Two-stroke engines, used in chainsaws and trimmers, complete the cycle in one crankshaft turn. They are lighter and make power every turn, but their oil is mixed into the fuel and burned, so they create more pollution.

Check your reading

1. List the four strokes in order.

_____

_____

2. What does the crankshaft do?

_____

_____

3. Give one difference between 2-stroke and 4-stroke engines.

_____

_____

🔩 Engine Parts ID: Lab Sheet

Small Engines Overhaul · Unit 3 · Level 3 · goes with Small Engine Parts ID Quiz + Engine Tools/Parts sheets

Name: _____

Question: Can you identify the main outside parts of a small engine and explain each part's job?

Materials

practice engine, label cards, parts sheet

⚠️ Safety

- Safety glasses
- Spark plug wire disconnected
- Tools with adult supervision

Prediction

Which part do you think is hardest to find?

_____

_____

Procedure

1. Disconnect plug wire.
2. Label 8 parts.
3. Write each part's job.
4. Name the tool to remove each.
5. Photo.

Data

Part	Job	Tool to remove
Spark plug		
Carburetor		
Flywheel		
Governor		

Analysis (Claim, Evidence, Reasoning)

1. Claim: which part is most important for starting the engine? Use evidence from the 4-stroke cycle.

⚙️ 4-Stroke Theory & Engine Parts: Quick Check

Small Engines Overhaul · Unit 3 · Level 3 (closer to the class work) · goes with the 4 Stroke Theory slides, Parts ID quiz and Engine Tools/Parts sheets

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. On the power stroke the piston moves…

A. Up | B. Sideways | C. Down

2. The crankshaft turns how many times per four strokes?

A. Two | B. One | C. Four

3. Which part opens the valves?

A. Spark plug | B. Muffler | C. Camshaft

4. A 2-stroke engine's oil is…

A. Mixed with the fuel | B. Only in the crankcase | C. Not needed

5. Explain what happens on the compression stroke.

6. Why does a single-cylinder engine need a flywheel?

⚙️ 4-Stroke Theory & Engine Parts: Guided Notes · ANSWER KEY

Small Engines Overhaul · Unit 3 · Level 3 (closer to the class work) · goes with the 4 Stroke Theory slides, Parts ID quiz and Engine Tools/Parts sheets

Name: _____

Use the class slides and your reading. Key words:

stroke *intake* *compression* *power* *exhaust* *piston* *connecting rod* *crankshaft* *camshaft* *flywheel* *two-stroke*

A. Strokes

1. Fill in the table.

Stroke	Piston direction	Valves
Intake	**down**	**intake open**
Compression	**up**	**both closed**
Power	**down**	**both closed**
Exhaust	**up**	**exhaust open**

B. Parts

2. The **connecting rod** links the piston to the crankshaft.
3. The **camshaft** opens the valves.
4. The **flywheel** keeps the engine turning between power strokes.

C. Compare

5. Why do chainsaws use 2-stroke engines?
Lighter and more power for their size; work at any angle.

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

🔩 Engine Parts ID: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 3 · Level 3 · goes with Small Engine Parts ID Quiz + Engine Tools/Parts sheets

Name: _____

Question: Can you identify the main outside parts of a small engine and explain each part's job?

Materials

practice engine, label cards, parts sheet

⚠️ Safety

- Safety glasses
- Spark plug wire disconnected
- Tools with adult supervision

Prediction

Which part do you think is hardest to find?
Often the governor.

Procedure

1. Disconnect plug wire.
2. Label 8 parts.
3. Write each part's job.
4. Name the tool to remove each.
5. Photo.

Data

Part	Job	Tool to remove
Spark plug	**makes the spark**	**spark plug socket**
Carburetor	**mixes air and fuel**	**socket/wrench**
Flywheel	**momentum, magnets**	**flywheel puller**
Governor	**controls speed**	**wrench/screwdriver**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: which part is most important for starting the engine? Use evidence from the 4-stroke cycle.
Spark plug (or carburetor): without spark there is no power stroke.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

⚙️ 4-Stroke Theory & Engine Parts: Quick Check · ANSWER KEY

Small Engines Overhaul · Unit 3 · Level 3 (closer to the class work) · goes with the 4 Stroke Theory slides, Parts ID quiz and Engine Tools/Parts sheets

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. On the power stroke the piston moves…

A. Up | B. Sideways | **C. Down**

2. The crankshaft turns how many times per four strokes?

A. Two | B. One | C. Four

3. Which part opens the valves?

A. Spark plug | B. Muffler | **C. Camshaft**

4. A 2-stroke engine's oil is…

A. Mixed with the fuel | B. Only in the crankcase | C. Not needed

5. Explain what happens on the compression stroke.

Both valves close; piston moves up and squeezes the air-fuel mix.

6. Why does a single-cylinder engine need a flywheel?

Its momentum carries the engine through the three non-power strokes.

🔑 *Written by our team from his unit calendar, worksheets and review games, please check.*